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Semiconductor package

Abstract

A semiconductor package includes an upper substrate having a first surface and a second surface which are opposite to each other, a semiconductor chip on the first surface of the upper substrate, a buffer layer on the second surface of the upper substrate, a mold layer between the second surface of the upper substrate and the buffer layer, a plurality of through-electrodes penetrating the upper substrate and the mold layer, an interconnection layer between the first surface of the upper substrate and the semiconductor chip and configured to electrically connect the semiconductor chip to the plurality of through-electrodes, and a plurality of bumps disposed on the buffer layer, spaced apart from the mold layer, and electrically connected to the plurality of through-electrodes. The mold layer includes an insulating material of which a coefficient of thermal expansion is greater than that of the upper substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This U.S. non-provisional patent application is a continuation of U.S. application Ser. No. 17/181,116, filed on Feb. 22, 2021, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2020-0085590, filed on Jul. 10, 2020, in the Korean Intellectual Property Office, the disclosure of each of which is hereby incorporated by reference in its entirety.

BACKGROUND

(1) Embodiments of the inventive concepts relate to a semiconductor package and, more particularly, to a semiconductor package in which a plurality of semiconductor chips are mounted.

(2) An integrated circuit chip may be realized in the form of a semiconductor package so as to be appropriately applied to an electronic product. In a typical semiconductor package, a semiconductor chip may be mounted on a printed circuit board (PCB) and may be electrically connected to the PCB through bonding wires or bumps. Small, lightweight and multi-functional electronic devices have been demanded with the development of the electronic industry, and thus a multi-chip package and/or a system-in package have been suggested. The multi-chip package may be a single semiconductor package in which chips are stacked, and the system-in package may be a single semiconductor package in which different kinds of chips are mounted to operate as a system.

SUMMARY

(3) Embodiments of the inventive concepts may provide a semiconductor package capable of easily controlling warpage.

(4) Embodiments of the inventive concepts may also provide a semiconductor package with excellent reliability.

(5) According to an exemplary embodiment of the present invention, a semiconductor package includes an upper substrate having a first surface and a second surface which are opposite to each other, a semiconductor chip on the first surface of the upper substrate, a buffer layer on the second surface of the upper substrate, a mold layer between the second surface of the upper substrate and the buffer layer, a plurality of through-electrodes penetrating the upper substrate and the mold layer, an interconnection layer between the first surface of the upper substrate and the semiconductor chip and configured to electrically connect the semiconductor chip to the plurality of through-electrodes, and a plurality of bumps disposed on the buffer layer, spaced apart from the mold layer, and electrically connected to the plurality of through-electrodes. The mold layer

includes an insulating material of which a coefficient of thermal expansion is greater than that of the upper substrate.

(6) According to an exemplary embodiment of the present invention, a semiconductor package includes an upper substrate having a first surface and a second surface which are opposite to each other, a plurality of semiconductor chips on the first surface of the upper substrate, a mold layer on the second surface of the upper substrate, the mold layer including a material of which a coefficient of thermal expansion is greater than that of the upper substrate, a plurality of first through-electrodes penetrating the upper substrate, and a plurality of second through-electrodes penetrating the mold layer. Each of the plurality of first through-electrodes is connected to a corresponding one of the plurality of second through-electrodes. Each of the first plurality of through-electrodes has a first width in a first direction parallel to the first surface of the upper substrate. Each of the second plurality of through-electrodes has a second width in the first direction, the second width being greater than the first width.

(7) According to an exemplary embodiment of the present invention, a semiconductor package includes an upper substrate on a lower substrate, a plurality of semiconductor chips mounted on the upper substrate, an interconnection layer between the upper substrate and the plurality of semiconductor chips, the plurality of semiconductor chips being connected to a plurality of metal interconnection lines in the interconnection layer, a mold layer between the upper substrate and the lower substrate, a buffer layer between the mold layer and the lower substrate, wherein the buffer layer includes a first surface that is in contact with the mold layer and a second surface opposite to the first surface of the buffer layer, a plurality of bumps disposed between the mold layer and the lower substrate and connected to the lower substrate, and a plurality of through-electrodes penetrating the upper substrate and the mold layer. Each of the plurality of bumps includes a head portion and a pillar portion. The pillar portion extends from the head portion into the buffer layer and is connected to a corresponding through-electrode of the plurality of through-electrodes. The head portion is connected to the pillar portion and is disposed on the second surface of the buffer layer. The mold layer includes an insulating material of which a coefficient of thermal expansion is greater than that of the upper substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The inventive concepts will become more apparent in view of the attached drawings and accompanying detailed description.

(2) FIG. 1 is a plan view illustrating a semiconductor package according to some embodiments of the inventive concepts.

(3) FIG. 2 is a cross-sectional view taken along a line I-I' of FIG. 1.

(4) FIGS. 3 to 8 are cross-sectional views corresponding to the line I-I' of FIG. 1 to illustrate a method of manufacturing a semiconductor package, according to some embodiments of the inventive concepts.

(5) FIG. 9 is a cross-sectional view corresponding to the line I-I' of FIG. 1 to illustrate a semiconductor package according to some embodiments of the inventive concepts.

(6) FIG. 10 is a cross-sectional view corresponding to the line I-I' of FIG. 1 to illustrate a method of manufacturing a semiconductor package, according to some embodiments of the inventive concepts.

(7) FIG. 11 is a cross-sectional view corresponding to the line I-I' of FIG. 1 to illustrate a semiconductor package according to some embodiments of the inventive concepts.

(8) FIG. 12 is a cross-sectional view corresponding to the line I-I' of FIG. 1 to illustrate a method of manufacturing a semiconductor package, according to some embodiments of the inventive

concepts.

(9) FIG. 13 is a cross-sectional view corresponding to the line I-I' of FIG. 1 to illustrate a semiconductor package according to some embodiments of the inventive concepts.

(10) FIGS. 14 to 16 are cross-sectional views corresponding to the line I-I' of FIG. 1 to illustrate a method of manufacturing a semiconductor package, according to some embodiments of the inventive concepts.

(11) FIG. 17 is a cross-sectional view corresponding to the line I-I' of FIG. 1 to illustrate a semiconductor package according to some embodiments of the inventive concepts.

(12) FIG. 18 is a cross-sectional view corresponding to the line I-I' of FIG. 1 to illustrate a method of manufacturing a semiconductor package, according to some embodiments of the inventive concepts.

(13) FIG. 19 is a cross-sectional view corresponding to the line I-I' of FIG. 1 to illustrate a semiconductor package according to some embodiments of the inventive concepts.

(14) FIG. 20 is a cross-sectional view illustrating a semiconductor package according to some embodiments of the inventive concepts.

(15) FIG. 21 is a cross-sectional view illustrating a semiconductor package according to some embodiments of the inventive concepts.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(16) Hereinafter, embodiments of the inventive concepts will be described in detail with reference to the accompanying drawings.

(17) FIG. 1 is a plan view illustrating a semiconductor package according to some embodiments of the inventive concepts. FIG. 2 is a cross-sectional view taken along a line I-I' of FIG. 1.

(18) Referring to FIGS. 1 and 2, a semiconductor package 1000 may include a lower substrate 100, an upper substrate 210 on the lower substrate 100, a mold layer 240 between the lower substrate 100 and the upper substrate 210, a buffer layer 260 between the lower substrate 100 and the mold layer 240, a plurality of semiconductor chips 300 disposed on the upper substrate 210, and an interconnection layer 220 disposed between the upper substrate 210 and the plurality of semiconductor chips 300.

(19) The upper substrate 210 may have a first surface 210a and a second surface 210b which are opposite to each other. The upper substrate 210 may be disposed on the lower substrate 100 in such a way that the second surface 210b of the upper substrate 210 faces a top surface 100U of the lower substrate 100. The plurality of semiconductor chips 300 may be disposed on the first surface 210a of the upper substrate 210, and the interconnection layer 220 may be disposed between the first surface 210a of the upper substrate 210 and the plurality of semiconductor chips 300. The buffer layer 260 may be disposed between the second surface 210b of the upper substrate 210 and the lower substrate 100, and the mold layer 240 may be disposed between the second surface 210b of the upper substrate 210 and the buffer layer 260. In an exemplary embodiment, the buffer layer 260 may include a first surface that is in contact with the mold layer 240 and a second surface that is opposite to the first surface of the buffer layer 260. It will be understood that when an element is referred to as being “connected” or “coupled” to or “on” another element, it can be directly connected or coupled to or on the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, or as “contacting” or “in contact with” another element, there are no intervening elements present at the point of contact.

(20) The upper substrate 210 and the interconnection layer 220 may be referred to as an interposer substrate 200. The interconnection layer 220 may include metal interconnection lines 222 disposed on the first surface 210a of the upper substrate 210, first conductive pads 224 on the metal interconnection lines 222, and an interconnection insulating layer 223 covering the metal interconnection lines 222 and the first conductive pads 224. Uppermost metal interconnection lines 222 of the metal interconnection lines 222 may be connected to the first conductive pads 224, and

the interconnection insulating layer **223** may expose top surfaces of the first conductive pads **224**.

(21) The upper substrate **210** may be, for example, a silicon substrate. A plurality of through-electrodes **250** may be disposed in the upper substrate **210** and may be connected to lowermost metal interconnection lines **222** of the metal interconnection lines **222**. Each of the plurality of through-electrodes **250** may penetrate the upper substrate **210** and may be connected to a corresponding one of the metal interconnection lines **222**. Each of the plurality of through-electrodes **250** may extend into the mold layer **240** to penetrate the mold layer **240**. The plurality of through-electrodes **250** may include a conductive material (e.g., copper (Cu)). An insulating liner **252** may be disposed between the upper substrate **210** and each of the plurality of through-electrodes **250**. In some embodiments, the insulating liner **252** may extend between the mold layer **240** and each of the plurality of through-electrodes **250**. The insulating liner **252** may include an insulating material.

(22) In some embodiments, an insulating layer **230** may be disposed between the second surface **210b** of the upper substrate **210** and the mold layer **240**. For example, the insulating layer **230** may include silicon oxide and/or silicon nitride. Each of the plurality of through-electrodes **250** may penetrate the insulating layer **230** and may extend into the mold layer **240**. The insulating liner **252** may extend between the insulating layer **230** and each of the plurality of through-electrodes **250**.

(23) The mold layer **240** may be or may include an insulating material of which a coefficient of thermal expansion is greater than that of the upper substrate **210**. The mold layer **240** may include fillers (e.g., silica). For example, the mold layer **240** may include an epoxy molding compound (EMC) including the fillers (e.g., silica). The mold layer **240** may cover a portion of a sidewall of each of the plurality of through-electrodes **250**. In some embodiments, the insulating liner **252** may be disposed between the mold layer **240** and the sidewall of each of the plurality of through-electrodes **250**.

(24) Each of the upper substrate **210** and the mold layer **240** may have a thickness in a first direction **D1** perpendicular to the first surface **210a** of the upper substrate **210**. A thickness **240T** of the mold layer **240** may be less than a thickness **210T** of the upper substrate **210**. For example, the thickness **210T** of the upper substrate **210** may have a value in a range from about 30 μm to about 110 μm , and the thickness **240T** of the mold layer **240** may have a value in a range from about 5 μm to about 100 μm . The insulating layer **230** may have a thickness **230T** in the first direction **D1**, and the thickness **230T** of the insulating layer **230** may be less than the thickness **240T** of the mold layer **240**. For example, the thickness **230T** of the insulating layer **230** may have a value in a range from about 1 μm to about 2 μm . The interconnection layer **220** may have a thickness in the first direction **D1**. For example, the thickness of the interconnection layer **220** may have a value in a range from about 10 μm to about 20 μm . Terms such as “about” or “approximately” may reflect amounts, sizes, orientations, or layouts that vary only in a small relative manner, and/or in a way that does not significantly alter the operation, functionality, or structure of certain elements. For example, a range from “about 0.1 to about 1” may encompass a range such as a 0%-5% deviation around 0.1 and a 0% to 5% deviation around 1, especially if such deviation maintains the same effect as the listed range.

(25) For example, the buffer layer **260** may include a photosensitive insulating material (e.g., a photo imagable dielectric (PID)). The buffer layer **260** may have a thickness **260T** in the first direction **D1**, and the thickness **260T** of the buffer layer **260** may be less than the thickness **240T** of the mold layer **240**. The thickness **230T** of the insulating layer **230** may be less than the thickness **260T** of the buffer layer **260**. For example, the thickness **260T** of the buffer layer **260** may have a value in a range from about 3 μm to about 7 μm .

(26) In some embodiments, second conductive pads **264** may be disposed between the mold layer **240** and the buffer layer **260**. In an exemplary embodiment, the second conductive pads **264** may be buried in the buffer layer **260** and adjacent to the first surface of the buffer layer **260**. Each of the plurality of through-electrodes **250** may penetrate the mold layer **240** so as to be connected to a

corresponding one of the second conductive pads **264**. In some embodiments, sizes (or a pitch) of the second conductive pads **264** may be greater than sizes (or a pitch) of the first conductive pads **224**. The buffer layer **260** may cover the second conductive pads **264** and may extend between the second conductive pads **264** so as to be in contact with the mold layer **240**.

(27) A plurality of bumps **270** may be disposed between the lower substrate **100** and the buffer layer **260**. The plurality of bumps **270** may be disposed on the buffer layer **260** and may be spaced apart from the mold layer **240**. In some embodiments, each of the plurality of bumps **270** may extend into the buffer layer **260** and may be connected to a corresponding one of the second conductive pads **264**. Each of the plurality of bumps **270** may be electrically connected to a corresponding one of the plurality of through-electrodes **250** through the corresponding second conductive pad **264**. Each of the plurality of bumps **270** may include a conductive material and may have at least one of a solder ball shape, a bump shape, or a pillar shape. In an exemplary embodiment, each of the plurality of bumps **270** may include a head portion and a pillar portion. The pillar portion of each bump **270** may extend from the head portion into the buffer layer **260** to be in contact with the corresponding second conductive pad **264**. The head portion of each bump **270** may be connected to the pillar portion thereof, and may be disposed on the second surface of the buffer layer **260**.

(28) The lower substrate **100** may include first lower substrate pads **110** adjacent to the top surface **100U** of the lower substrate **100**, and second lower substrate pads **120** adjacent to a bottom surface **100L** of the lower substrate **100**. The first and second lower substrate pads **110** and **120** may include a conductive material. In some embodiments, the lower substrate **100** may be a printed circuit board (PCB). External terminals **130** may be disposed on the bottom surface **100L** of the lower substrate **100** and may be connected to the second lower substrate pads **120**, respectively. The plurality of bumps **270** may be connected to corresponding ones of the first lower substrate pads **110**, respectively. The first lower substrate pads **110** may be electrically connected to the second lower substrate pads **120** through internal interconnection lines (not shown) provided in the lower substrate **100**.

(29) A lower underfill layer **280** may be disposed between the lower substrate **100** and the buffer layer **260** and may fill a space between the plurality of bumps **270**. The lower underfill layer **280** may include an insulating polymer material such as an epoxy resin.

(30) The plurality of semiconductor chips **300** may be disposed on the interconnection layer **220** on the first surface **210a** of the upper substrate **210**. The plurality of semiconductor chips **300** may be horizontally spaced apart from each other on the interconnection layer **220** in a second direction **D2** parallel to the first surface **210a** of the upper substrate **210**. A first surface **300S** of each of the plurality of semiconductor chips **300** may face the first surface **210a** of the upper substrate **210**. Each of the plurality of semiconductor chips **300** may include a circuit layer **310** adjacent to the first surface **300S** of each of the plurality of semiconductor chips **300**. The circuit layer **310** may include a plurality of transistors to form integrated circuits.

(31) Each of the plurality of semiconductor chips **300** may include chip pads **320** adjacent to the first surface **300S** of each of the plurality of semiconductor chips **300**. The chip pads **320** may include a conductive material. The chip pads **320** may be connected to corresponding ones of the first conductive pads **224**. In some embodiments, connection bumps **330** may be disposed between the chip pads **320** and the first conductive pads **224**. Each of the chip pads **320** may be connected to a corresponding first conductive pad **224** through a corresponding one of the connection bumps **330**. Each of the connection bumps **330** may include a conductive material and may have at least one of a solder ball shape, a bump shape, or a pillar shape.

(32) In some embodiments, an upper underfill layer **340** may be disposed between the interconnection layer **220** and the first surface **300S** of each of the plurality of semiconductor chips **300**. The upper underfill layer **340** may be locally disposed on the first surface **300S** of each of the plurality of semiconductor chips **300** and may fill a space between the connection bumps **330**. The

upper underfill layer **340** may include an insulating polymer material such as an epoxy resin. In certain embodiments, the connection bumps **330** and the upper underfill layer **340** may be omitted unlike FIG. **2**. The chip pads **320** may be in contact with and bonded directly to the corresponding first conductive pads **224**.

(33) In some embodiments, the plurality of semiconductor chips **300** may be the same kind of semiconductor chips. For example, the plurality of semiconductor chips **300** may correspond to memory chips, logic chips, application processor (AP) chips, or system-on-chips (SOCs). In certain embodiments, the plurality of semiconductor chips **300** may include different kinds of semiconductor chips from each other. For example, the plurality of semiconductor chips **300** may include different semiconductor chips selected from a group consisting of a memory chip, a logic chip, an application processor (AP) chip, and a system-on-chip (SOC). In certain embodiments, at least one of the plurality of semiconductor chips **300** may be a high bandwidth memory (HBM) chip which includes a plurality of memory chips stacked in the first direction **D1** perpendicular to the first surface **210a** of the upper substrate **210**.

(34) Each of the plurality of semiconductor chips **300** may be electrically connected to the metal interconnection lines **222** through the chip pads **320**, corresponding connection bumps **330** and corresponding first conductive pads **224**. Each of the plurality of semiconductor chips **300** may be electrically connected to the lower substrate **100** through corresponding metal interconnection lines **222** and corresponding through-electrodes **250**. The plurality of semiconductor chips **300** may be electrically connected to each other through the metal interconnection lines **222** in the interconnection layer **220**.

(35) When the metal interconnection lines **222** are disposed in the interconnection layer **220** and the through-electrodes **250** are disposed in the upper substrate **210**, warpage of the interposer substrate **200** may occur by structural imbalance between the metal interconnection lines **222** and the through-electrodes **250**. For example, a distribution pattern of the through-electrodes **250** and a distribution pattern of the metal interconnection lines **222** may be different from each other, and such difference in the distribution patterns may cause warpage of the interposer substrate **200**.

(36) According to the embodiments of the inventive concepts, the interconnection layer **220** may be disposed on the first surface **210a** of the upper substrate **210**, and the mold layer **240** may be disposed on the second surface **210b** of the upper substrate **210**. The mold layer **240** may include the insulating material of which the coefficient of thermal expansion is greater than that of the upper substrate **210**, and thus the warpage of the interposer substrate **200** may be reduced. The mold layer **240** may include the fillers (e.g., silica). The coefficient of thermal expansion of the mold layer **240** may be adjusted according to the amount of the fillers in the mold layer **240**. Therefore, the warpage of the interposer substrate **200** may be easily adjusted.

(37) Furthermore, the buffer layer **260** may be disposed on the second surface **210b** of the upper substrate **210**, and the mold layer **240** may be disposed between the second surface **210b** of the upper substrate **210** and the buffer layer **260**. The plurality of bumps **270** may be disposed on the buffer layer **260** so as to be spaced apart from the mold layer **240**. The buffer layer **260** may reduce stress applied to the plurality of bumps **270**.

(38) As a result, the warpage may be easily controlled, and a semiconductor package with excellent reliability may be provided.

(39) FIGS. **3** to **8** are cross-sectional views corresponding to the line I-I' of FIG. **1** to illustrate a method of manufacturing a semiconductor package, according to some embodiments of the inventive concepts. Hereinafter, the descriptions to the same features as in the semiconductor package **1000** mentioned with reference to FIGS. **1** and **2** will be omitted for the purpose of ease and convenience in explanation.

(40) Referring to FIG. **3**, a plurality of through-electrodes **250** and a plurality of insulating liners **252** may be formed in an upper substrate **210**. For example, the formation of the plurality of through-electrodes **250** and the plurality of insulating liners **252** may include forming through-

holes **250H** penetrating a portion of the upper substrate **210**, forming an insulating liner layer filling a portion of each of the through-holes **250H** on a first surface **210a** of the upper substrate **210**, forming an electrode layer filling a remaining portion of each of the through-holes **250H** on the insulating liner layer, and planarizing the electrode layer and the insulating liner layer to expose the first surface **210a** of the upper substrate **210**. Each of the through-holes **250H** may extend from the first surface **210a** of the upper substrate **210** toward a preliminary second surface **210b-P** of the upper substrate **210** and may be spaced apart from the second surface **210b** of the upper substrate **210**. The plurality of through-electrodes **250** and the plurality of insulating liners **252** may be locally formed in the through-holes **250H** by the planarization process. Each of the plurality of insulating liners **252** may be disposed between the upper substrate **210** and each of the plurality of through-electrodes **250**.

(41) Referring to FIG. 4, an interconnection layer **220** may be formed on the first surface **210a** of the upper substrate **210**. The formation of the interconnection layer **220** may include forming metal interconnection lines **222** on the first surface **210a** of the upper substrate **210**, forming first conductive pads **224** on the metal interconnection lines **222**, and forming an interconnection insulating layer **223** covering the metal interconnection lines **222** and the first conductive pads **224**.

(42) A first etching process may be performed on the preliminary second surface **210b-P** of the upper substrate **210**. A portion of the upper substrate **210** may be selectively etched by the first etching process, and thus a portion of each of the plurality of through-electrodes **250** and a portion of each of the plurality of insulating liners **252** may protrude from a second surface **210b** of the upper substrate **210**. In some embodiments, a preliminary insulating layer **230-P** may be formed on the second surface **210b** of the upper substrate **210** and may cover the protruding portion of each of the plurality of through-electrodes **250** and the protruding portion of each of the plurality of insulating liners **252**. The insulating layer **230** may be formed using, for example, a chemical vapor deposition (CVD) process.

(43) Referring to FIG. 5, a second etching process may be performed on the second surface **210b** of the upper substrate **210**. A portion of the preliminary insulating layer **230-P** may be selectively etched by the second etching process, and thus a portion of each of the plurality of through-electrodes **250** and a portion of each of the plurality of insulating liners **252** may protrude from an insulating layer **230**. In certain embodiments, the formation of the insulating layer **230** and the second etching process may be omitted unlike the drawings.

(44) A preliminary mold layer **240-P** may be formed on the second surface **210b** of the upper substrate **210** and may cover the insulating layer **230**, the protruding portion of each of the plurality of through-electrodes **250**, and the protruding portion of each of the plurality of insulating liners **252**. The mold layer **240** may be formed of an insulating material having a greater coefficient of thermal expansion than that of the upper substrate **210** and may include fillers (e.g., silica).

(45) Referring to FIG. 6, a grinding process may be performed on the preliminary mold layer **240-P**. A portion of the preliminary mold layer **240-P**, a portion of each of the plurality of through-electrodes **250** and a portion of each of the plurality of insulating liners **252** may be removed by the grinding process. Thus, an end of each of the plurality of through-electrodes **250** and an end of each of the plurality of insulating liners **252** may be exposed.

(46) Referring to FIG. 7, according to some embodiments, second conductive pads **264** may be formed on the ground surface of the mold layer **240**. Each of the second conductive pads **264** may cover the end of each of the plurality of through-electrodes **250**. A buffer layer **260** may be formed on the ground surface of the mold layer **240** and may cover the second conductive pads **264**. The buffer layer **260** may extend between the second conductive pads **264** so as to be in contact with the mold layer **240**.

(47) A plurality of bumps **270** may be formed on the buffer layer **260** on the second surface **210b** of the upper substrate **210**. For example, the formation of the plurality of bumps **270** may include removing a portion of the buffer layer **260** to form a recess exposing a portion of each of the

second conductive pads **264**. Each of the plurality of bumps **270** may be formed to fill the recess and may be connected to a corresponding second conductive pad **264**.

(48) The upper substrate **210**, the interconnection layer **220**, the insulating layer **230**, the mold layer **240**, the buffer layer **260**, the plurality of through-electrodes **250**, the plurality of insulating liners **252**, the second conductive pads **264** and the plurality of bumps **270** may constitute a stack structure SS.

(49) Referring to FIG. **8**, the stack structure SS may be mounted on a lower substrate **100**. The lower substrate **100** may include first lower substrate pads **110** adjacent to a top surface **100U** of the lower substrate **100**, and second lower substrate pads **120** adjacent to a bottom surface **100L** of the lower substrate **100**. The stack structure SS may be disposed on the lower substrate **100** in such a way that the second surface **210b** of the upper substrate **210** faces the top surface **100U** of the lower substrate **100**. The plurality of bumps **270** may be connected to corresponding ones of the first lower substrate pads **110**. In an exemplary embodiment, each bump **270** may be connected to a corresponding one of the first lower substrate pads **110**.

(50) A lower underfill layer **280** may be formed between the lower substrate **100** and the buffer layer **260**. The lower underfill layer **280** may be formed to fill a space between the plurality of bumps **270**. External terminals **130** may be disposed on the bottom surface **100L** of the lower substrate **100** and may be connected to the second lower substrate pads **120**, respectively.

(51) Referring again to FIGS. **1** and **2**, connection bumps **330** may be formed on the first conductive pads **224**. Each of the connection bumps **330** may be formed on a corresponding one of the first conductive pads **224** and may be connected to the corresponding first conductive pad **224**.

(52) A plurality of semiconductor chips **300** may be disposed on the first surface **210a** of the upper substrate **210** and may be mounted on the interconnection layer **220**. Each of the plurality of semiconductor chips **300** may include chip pads **320**, and the chip pads **320** may be disposed adjacent to the first surface **300S** of each of the plurality of semiconductor chips **300**. Each of the plurality of semiconductor chips **300** may be provided on the interconnection layer **220** in such a way that the chip pads **320** are in contact with corresponding ones of the connection bumps **330**. In an exemplary embodiment, each chip pad **320** may be in contact with a corresponding one of the connection bumps **330**. Each of the plurality of semiconductor chips **300** may include a circuit layer **310** adjacent to the first surface **300S** of each of the plurality of semiconductor chips **300**. The plurality of semiconductor chips **300** may be horizontally spaced apart from each other on the interconnection layer **220**.

(53) An upper underfill layer **340** may be formed between the interconnection layer **220** and the first surface **300S** of each of the plurality of semiconductor chips **300**. The upper underfill layer **340** may be formed to fill a space between the connection bumps **330**.

(54) FIG. **9** is a cross-sectional view corresponding to the line I-I' of FIG. **1** to illustrate a semiconductor package **1100** according to some embodiments of the inventive concepts. Hereinafter, differences between the semiconductor package **1100** and the semiconductor package **1000** described with reference to FIGS. **1** and **2** will be mainly described for the purpose of ease and convenience in explanation.

(55) Referring to FIG. **9**, according to some embodiments, a redistribution layer **290** may be disposed between the mold layer **240** and the buffer layer **260**. The redistribution layer **290** may include redistribution patterns **292** and a redistribution insulating layer **293** covering the redistribution patterns **292**. The redistribution insulating layer **293** may extend between the redistribution patterns **292** so as to be in contact with the mold layer **240**. The redistribution patterns **292** may include a conductive material, and the redistribution insulating layer **293** may include an insulating material. Each of the plurality of through-electrodes **250** may penetrate the mold layer **240** so as to be connected to a corresponding one of the redistribution patterns **292**. The buffer layer **260** may be spaced apart from the mold layer **240** with the redistribution layer **290** interposed therebetween.

(56) The plurality of bumps **270** may be disposed between the lower substrate **100** and the buffer layer **260**. The plurality of bumps **270** may be disposed on the buffer layer **260** and may be spaced apart from the mold layer **240**. In some embodiments, each of the plurality of bumps **270** may extend into the buffer layer **260** and may be connected to a corresponding one of the redistribution patterns **292**. Each of the plurality of bumps **270** may be electrically connected to a corresponding one of the plurality of through-electrodes **250** through the corresponding redistribution pattern **292**.

(57) FIG. **10** is a cross-sectional view corresponding to the line I-I' of FIG. **1** to illustrate a method of manufacturing a semiconductor package, according to some embodiments of the inventive concepts. Hereinafter, differences between the present embodiments and the above embodiments described with reference to FIGS. **1** to **8** will be mainly described for the purpose of ease and convenience in explanation.

(58) As described with reference to FIG. **6**, the grinding process may be performed on the preliminary mold layer **240-P** as shown in FIG. **5**, and thus a portion of the preliminary mold layer **240-P**, a portion of each of the plurality of through-electrodes **250** and a portion of each of the plurality of insulating liners **252** may be removed. Therefore, an end of each of the plurality of through-electrodes **250** and an end of each of the plurality of insulating liners **252** may be exposed.

(59) Referring to FIG. **10**, a redistribution layer **290** may be formed on the ground surface of the mold layer **240**. The formation of the redistribution layer **290** may include forming redistribution patterns **292** on the ground surface of the mold layer **240**, and forming a redistribution insulating layer **293** covering the redistribution patterns **292** on the ground surface of the mold layer **240**. The redistribution patterns **292** may be formed to be connected to the plurality of through-electrodes **250**, respectively. The redistribution insulating layer **293** may extend between the redistribution patterns **292** so as to be in contact with the mold layer **240**.

(60) A buffer layer **260** may be formed on the redistribution layer **290** on the second surface **210b** of the upper substrate **210**. A plurality of bumps **270** may be formed on the buffer layer **260** on the second surface **210b** of the upper substrate **210**. For example, the formation of the plurality of bumps **270** may include removing a portion of the buffer layer **260** to form a recess exposing a corresponding one of the redistribution patterns **292**. Each of the plurality of bumps **270** may be formed to fill the recess and may be connected to the corresponding redistribution pattern **292**.

(61) Except for the differences as described above, the method of manufacturing the semiconductor package according to the present embodiments may be substantially the same as the method of manufacturing the semiconductor package described with reference to FIGS. **1** to **8**.

(62) FIG. **11** is a cross-sectional view corresponding to the line I-I' of FIG. **1** to illustrate a semiconductor package **1200** according to some embodiments of the inventive concepts. Hereinafter, differences between the semiconductor package **1200** and the semiconductor package **1000** described with reference to FIGS. **1** and **2** will be mainly described for the purpose of ease and convenience in explanation.

(63) Referring to FIG. **11**, according to some embodiments, the second conductive pads **264** may be omitted. The plurality of bumps **270** may be disposed on the buffer layer **260**, and each of the plurality of bumps **270** may penetrate the buffer layer **260** so as to be connected directly to a corresponding one of the plurality of through-electrodes **250**. Each of the plurality of bumps **270** may be in contact with the corresponding through-electrode **250**. The buffer layer **260** may be in contact with the mold layer **240** between the plurality of bumps **270**.

(64) FIG. **12** is a cross-sectional view corresponding to the line I-I' of FIG. **1** to illustrate a method of manufacturing a semiconductor package, according to some embodiments of the inventive concepts. Hereinafter, differences between the present embodiments and the above embodiments described with reference to FIGS. **1** to **8** will be mainly described for the purpose of ease and convenience in explanation.

(65) As described with reference to FIG. **6**, the grinding process may be performed on the preliminary mold layer **240-P** as shown in FIG. **5**, and thus a portion of the preliminary mold layer

240-P, a portion of each of the plurality of through-electrodes **250** and a portion of each of the plurality of insulating liners **252** may be removed. Therefore, an end of each of the plurality of through-electrodes **250** and an end of each of the plurality of insulating liners **252** may be exposed. (66) Referring to FIG. **12**, a buffer layer **260** may be formed on the ground surface of the mold layer **240**. The buffer layer **260** may cover the end of each of the plurality of through-electrodes **250** and the end of each of the plurality of insulating liners **252**. A plurality of bumps **270** may be formed on the buffer layer **260**. For example, the formation of the plurality of bumps **270** may include removing a portion of the buffer layer **260** to form a recess exposing the end of each of the plurality of through-electrodes **250**. Each of the plurality of bumps **270** may be formed to fill the recess and may be connected directly to a corresponding through-electrode **250**.

(67) Except for the differences as described above, the method of manufacturing the semiconductor package according to the present embodiments may be substantially the same as the method of manufacturing the semiconductor package described with reference to FIGS. **1** to **8**.

(68) FIG. **13** is a cross-sectional view corresponding to the line I-I' of FIG. **1** to illustrate a semiconductor package **1300** according to some embodiments of the inventive concepts. Hereinafter, differences between the semiconductor package **1300** and the semiconductor package **1000** described with reference to FIGS. **1** and **2** will be mainly described for the purpose of ease and convenience in explanation.

(69) Referring to FIG. **13**, each of the plurality of through-electrodes **250** may include a first through-electrode **250a** penetrating the upper substrate **210**, and a second through-electrode **250b** penetrating the mold layer **240**. The first through-electrode **250a** may be connected to a lowermost metal interconnection line **222** of the metal interconnection lines **222** and may be connected to the second through-electrode **250b**. In some embodiments, the first through-electrode **250a** may penetrate the insulating layer **230** so as to be connected directly to the second through-electrode **250b**. It will be understood that, although the terms first, second, third etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. Unless the context indicates otherwise, these terms are only used to distinguish one element, component, region, layer or section from another element, component, region, layer or section, for example as a naming convention. Thus, a first element, component, region, layer or section discussed below in one section of the specification could be termed a second element, component, region, layer or section in another section of the specification or in the claims without departing from the teachings of the present invention. In addition, in certain cases, even if a term is not described using "first," "second," etc., in the specification, it may still be referred to as "first" or "second" in a claim in order to distinguish different claimed elements from each other.

(70) The insulating liner **252** may be disposed between the first through-electrode **250a** and the upper substrate **210**. In some embodiments, the insulating liner **252** may extend between the first through-electrode **250a** and the insulating layer **230**. The first through-electrode **250a** may include a conductive material (e.g., copper (Cu)). For example, the second through-electrode **250b** may include the same conductive material as the first through-electrode **250a**. Alternatively, the second through-electrode **250b** may include a different conductive material from that of the first through-electrode **250a**.

(71) Each of the first and second through-electrodes **250a** and **250b** may have a width in the second direction **D2** parallel to the first surface **210a** of the upper substrate **210**. A first width **W1** of the first through-electrode **250a** may be less than a second width **W2** of the second through-electrode **250b** (i.e., $W1 < W2$). The first through-electrode **250a** and the second through-electrode **250b** may overlap with each other in the first direction **D1** perpendicular to the first surface **210a** of the upper substrate **210**. The second through-electrode **250b** may be in contact with the first through-electrode **250a** and may be in contact with a bottommost surface of the insulating liner **252**. The mold layer **240** may be in contact with a sidewall of the second through-electrode **250b**.

(72) The second through-electrode **250b** may penetrate the mold layer **240** so as to be connected to a corresponding one of the second conductive pads **264**. The second through-electrode **250b** may be electrically connected to a corresponding one of the plurality of bumps **270** through the corresponding second conductive pad **264**.

(73) FIGS. **14** to **16** are cross-sectional views corresponding to the line I-I' of FIG. **1** to illustrate a method of manufacturing a semiconductor package, according to some embodiments of the inventive concepts. Hereinafter, differences between the present embodiments and the above embodiments described with reference to FIGS. **1** to **8** will be mainly described for the purpose of ease and convenience in explanation.

(74) Referring to FIG. **14**, a plurality of first through-electrodes **250a** and a plurality of insulating liners **252** may be formed in an upper substrate **210**. An interconnection layer **220** may be formed on a first surface **210a** of the upper substrate **210**, and an insulating layer **230** may be formed on a second surface **210b** of the upper substrate **210**. The plurality of first through-electrodes **250a**, the plurality of insulating liners **252**, the interconnection layer **220** and the insulating layer **230** may be formed by substantially the same method as of forming the plurality of through-electrodes **250**, the plurality of insulating liners **252**, the interconnection layer **220** and the insulating layer **230**, described with reference to FIGS. **3** and **4**.

(75) A grinding process may be performed on the insulating layer **230**, and a portion of the insulating layer **230**, a portion of each of the plurality of first through-electrodes **250a** and a portion of each of the plurality of insulating liners **252** may be removed by the grinding process. Thus, an end of each of the plurality of first through-electrodes **250a** and an end of each of the plurality of insulating liners **252** may be exposed.

(76) Referring to FIG. **15**, a plurality of second through-electrodes **250b** may be formed on a ground surface of the insulating layer **230**. For example, the formation of the plurality of second through-electrodes **250b** may include depositing a conductive layer on the insulating layer **230**, and patterning the conductive layer. A mold layer **240** may be formed on the ground surface of the insulating layer **230** and may cover the plurality of second through-electrodes **250b**. In some embodiments, an additional grinding process may be performed on the mold layer **240**, and ends of the plurality of second through-electrodes **250b** may be exposed by the additional grinding process.

(77) Referring to FIG. **16**, second conductive pads **264** may be formed on the mold layer **240**, and each of the second conductive pads **264** may cover the end of a corresponding one of the plurality of second through-electrodes **250b**. In an exemplary embodiment, each of the second conductive pads **264** may be in contact with the end of a corresponding one of the plurality of second through-electrodes **250b**. A buffer layer **260** may be formed on the mold layer **240** and may cover the second conductive pads **264**. A plurality of bumps **270** may be formed on the buffer layer **260**. Each of the plurality of bumps **270** may extend into the buffer layer **260** so as to be connected to a corresponding one of the second conductive pads **264**.

(78) Except for the differences as described above, the method of manufacturing the semiconductor package according to the present embodiments may be substantially the same as the method of manufacturing the semiconductor package described with reference to FIGS. **1** to **8**.

(79) FIG. **17** is a cross-sectional view corresponding to the line I-I' of FIG. **1** to illustrate a semiconductor package **1400** according to some embodiments of the inventive concepts. Hereinafter, differences between the semiconductor package **1400** and the semiconductor package **1000** described with reference to FIGS. **1** and **2** will be mainly described for the purpose of ease and convenience in explanation.

(80) Referring to FIG. **17**, the semiconductor package **1400** may further include an upper mold layer **350** disposed on the first surface **210a** of the upper substrate **210**. The upper mold layer **350** may be disposed on the interconnection layer **220** and may cover the plurality of semiconductor chips **300**. In some embodiments, the upper mold layer **350** may include the same material as the mold layer **240**. In certain embodiments, the upper mold layer **350** may include a different material

from that of the mold layer **240**. The upper mold layer **350** may include an insulating material (e.g., an epoxy molding compound).

(81) FIG. **18** is a cross-sectional view corresponding to the line I-I' of FIG. **1** to illustrate a method of manufacturing a semiconductor package, according to some embodiments of the inventive concepts. Hereinafter, differences between the present embodiments and the above embodiments described with reference to FIGS. **1** to **8** will be mainly described for the purpose of ease and convenience in explanation.

(82) The upper substrate **210**, the interconnection layer **220**, the insulating layer **230**, the mold layer **240**, the buffer layer **260**, the plurality of through-electrodes **250**, the plurality of insulating liners **252**, the second conductive pads **264** and the plurality of bumps **270** may be formed as described with reference to FIGS. **3** to **7**. The upper substrate **210**, the interconnection layer **220**, the insulating layer **230**, the mold layer **240**, the buffer layer **260**, the plurality of through-electrodes **250**, the plurality of insulating liners **252**, the second conductive pads **264** and the plurality of bumps **270** may constitute a stack structure SS.

(83) Referring to FIG. **18**, the stack structure SS may be provided on a carrier substrate CA. The stack structure SS may be provided on the carrier substrate CA in such a way that the second surface **210b** of the upper substrate **210** faces the carrier substrate CA.

(84) Connection bumps **330** may be formed on the first conductive pads **224**, and each of the connection bumps **330** may be connected to a corresponding one of the first conductive pads **224**.

(85) A plurality of semiconductor chips **300** may be disposed on the first surface **210a** of the upper substrate **210** and may be mounted on the interconnection layer **220**. Each of the plurality of semiconductor chips **300** may include chip pads **320** disposed adjacent to the first surface **300S** of each of the plurality of semiconductor chips **300**. Each of the plurality of semiconductor chips **300** may be provided on the interconnection layer **220** in such a way that the chip pads **320** are in contact with corresponding ones of the connection bumps **330**. In an exemplary embodiment, each chip pad **320** may be in contact with a corresponding one of the connection bumps **330**. The plurality of semiconductor chips **300** may be horizontally spaced apart from each other on the interconnection layer **220**.

(86) An upper underfill layer **340** may be formed between the interconnection layer **220** and the first surface **300S** of each of the plurality of semiconductor chips **300** and may fill a space between the connection bumps **330**. An upper mold layer **350** may be formed on the interconnection layer **220** to cover the plurality of semiconductor chips **300**.

(87) Referring again to FIG. **17**, the carrier substrate CA may be removed from the stack structure SS. Thereafter, the stack structure SS on which the plurality of semiconductor chips **300** is mounted may be mounted on a lower substrate **100**.

(88) Except for the differences as described above, the method of manufacturing the semiconductor package according to the present embodiments may be substantially the same as the method of manufacturing the semiconductor package described with reference to FIGS. **1** to **8**.

(89) FIG. **19** is a cross-sectional view corresponding to the line I-I' of FIG. **1** to illustrate a semiconductor package **1500** according to some embodiments of the inventive concepts. Hereinafter, differences between the semiconductor package **1500** and the semiconductor package **1000** described with reference to FIGS. **1** and **2** will be mainly described for the purpose of ease and convenience in explanation.

(90) Referring to FIG. **19**, the semiconductor package **1500** may further include a heat dissipation structure **500** disposed on the lower substrate **100**. The heat dissipation structure **500** may be disposed on the top surface **100U** of the lower substrate **100**, and the components of the semiconductor package **1000** described with reference to FIGS. **1** and **2** may be disposed inside the heat dissipation structure **500**. In an exemplary embodiment, a space may be formed between the heat dissipation structure and the top surface **100U** of the lower substrate **100**, and the components of the semiconductor package **1000** described with reference to FIGS. **1** and **2** may be disposed in

the space. The heat dissipation structure **500** may include a heat-conductive material. The heat-conductive material may include metal (e.g., copper and/or aluminum) or a carbon-containing material (e.g., graphene, graphite, and/or carbon nanotube). For example, the heat dissipation structure **500** may include a single metal layer or a plurality of stacked metal layers. The present invention is not limited thereto. In an exemplary embodiment, the heat dissipation structure **500** may include a heat sink or a heat pipe. For still other examples, the heat dissipation structure **500** may use a water cooling method.

(91) The semiconductor package **1500** may further include a heat-conductive layer **510** disposed between the heat dissipation structure **500** and each of the plurality of semiconductor chips **300**. The heat-conductive layer **510** may include a thermal interface material (TIM). The thermal interface material (TIM) may include, for example, a polymer and heat-conductive particles. The heat-conductive particles may be dispersed in the polymer. Heat generated from the plurality of semiconductor chips **300** may be transferred to the heat dissipation structure **500** through the heat-conductive layer **510**.

(92) FIG. **20** is a cross-sectional view illustrating a semiconductor package **1600** according to some embodiments of the inventive concepts.

(93) Referring to FIG. **20**, the semiconductor package **1600** may include a lower substrate **100**, a lower semiconductor chip **600** on the lower substrate **100**, a mold layer **240** between the lower substrate **100** and the lower semiconductor chip **600**, and a plurality of upper semiconductor chips **700** mounted on the lower semiconductor chip **600**. The plurality of upper semiconductor chips **700** may be vertically stacked on each other.

(94) The lower semiconductor chip **600** may have a first surface **600a** and a second surface **600b** which are opposite to each other. The lower semiconductor chip **600** may be disposed on the lower substrate **100** in such a way that the second surface **600b** faces a top surface **100U** of the lower substrate **100**. The mold layer **240** may be disposed between the second surface **600b** of the lower semiconductor chip **600** and the lower substrate **100**. The plurality of upper semiconductor chips **700** may be mounted on the first surface **600a** of the lower semiconductor chip **600**. The plurality of upper semiconductor chips **700** may be stacked in a first direction **D1** perpendicular to the first surface **600a** of the lower semiconductor chip **600**. The lower semiconductor chip **600** may include a circuit layer (not shown) adjacent to the first surface **600a** or the second surface **600b**. For example, the lower semiconductor chip **600** may be a logic chip, and the plurality of upper semiconductor chips **700** may be the same kind of a semiconductor device such as memory chips. In the present specification, the lower semiconductor chip **600** may serve as an upper substrate of FIGS. **1** and **2**.

(95) A plurality of through-electrodes **250** may be disposed in the lower semiconductor chip **600**. Each of the plurality of through-electrodes **250** may penetrate the lower semiconductor chip **600** and may extend into the mold layer **240** to penetrate the mold layer **240**. Each of the plurality of through-electrodes **250** may include a first through-electrode **250a** penetrating the lower semiconductor chip **600**, and a second through-electrode **250b** penetrating the mold layer **240**. In some embodiments, the first through-electrode **250a** may be connected directly to the second through-electrode **250b**. The first through-electrode **250a** may include a conductive material (e.g., copper (Cu)). For example, the second through-electrode **250b** may include the same conductive material as the first through-electrode **250a**. Alternatively, the second through-electrode **250b** may include a different conductive material from that of the first through-electrode **250a**.

(96) Each of the first and second through-electrodes **250a** and **250b** may have a width in a second direction **D2** parallel to the first surface **600a** of the lower semiconductor chip **600**. A first width **W1** of the first through-electrode **250a** may be less than a second width **W2** of the second through-electrode **250b** (i.e., $W1 < W2$). The first through-electrode **250a** and the second through-electrode **250b** may overlap with each other in the first direction **D1** perpendicular to the first surface **600a** of the lower semiconductor chip **600**. The second through-electrode **250b** may be in contact with the

first through-electrode **250a**, and the mold layer **240** may be in contact with a sidewall of the second through-electrode **250b**.

(97) The mold layer **240** may include an insulating material of which a coefficient of thermal expansion is greater than that of the lower semiconductor chip **600**. The mold layer **240** may include fillers (e.g., silica). For example, the mold layer **240** may include an epoxy molding compound (EMC) including the fillers (e.g., silica).

(98) Each of the lower semiconductor chip **600** and the mold layer **240** may have a thickness in the first direction **D1**. A thickness **240T** of the mold layer **240** may be less than a thickness **600T** of the lower semiconductor chip **600**. For example, the thickness **240T** of the mold layer **240** may have a value in a range from about 5 μm to about 100 μm .

(99) A plurality of bumps **270** may be disposed between the lower substrate **100** and the mold layer **240**. Each of the plurality of through-electrodes **250** may penetrate the mold layer **240** so as to be connected to a corresponding one of the plurality of bumps **270**. For example, each of the second through-electrodes **250b** may penetrate the mold layer **240** so as to be connected to a corresponding one of the plurality of bumps **270**. Each of the plurality of bumps **270** may include a conductive material and may have at least one of a solder ball shape, a bump shape, or a pillar shape.

(100) The lower substrate **100** may include first lower substrate pads **110** adjacent to the top surface **100U** of the lower substrate **100**, and second lower substrate pads **120** adjacent to a bottom surface **100L** of the lower substrate **100**. External terminals **130** may be disposed on the bottom surface **100L** of the lower substrate **100** and may be connected to the second lower substrate pads **120**, respectively. The plurality of bumps **270** may be connected to the first lower substrate pads **110**, respectively. The lower semiconductor chip **600** may be electrically connected to the lower substrate **100** through the plurality of through-electrodes **250** and the plurality of bumps **270**. The lower substrate **100** may be substantially the same as the lower substrate **100** described with reference to FIGS. **1** and **2**.

(101) A lower underfill layer **280** may be disposed between the lower substrate **100** and the mold layer **240** and may fill a space between the plurality of bumps **270**. The lower underfill layer **280** may include an insulating polymer material such as an epoxy resin.

(102) The plurality of upper semiconductor chips **700** may be electrically connected to each other through through-vias **710** and upper bumps **720**. In an exemplary embodiment, no through-vias are disposed in the uppermost upper semiconductor chip **700**, and the through-vias **710** may be disposed in each of the remaining upper semiconductor chips **700**. The upper bumps **720** may be disposed between the plurality of upper semiconductor chips **700** and between the lower semiconductor chip **600** and the lowermost one of the plurality of upper semiconductor chips **700**. The upper bumps **720** may be connected to the through-vias **710**.

(103) The plurality of through-electrodes **250** may be connected to corresponding ones of the upper bumps **720** disposed between the lowermost upper semiconductor chip **700** and the lower semiconductor chip **600**. For example, each of the first through-electrodes **250a** may be connected to a corresponding one of the upper bumps **720**. Thus, the lowermost upper semiconductor chip **700** may be electrically connected to the lower semiconductor chip **600**. The plurality of upper semiconductor chips **700** may be electrically connected to the lower substrate **100** through the through-vias **710**, the upper bumps **720**, the plurality of through-electrodes **250** and the plurality of bumps **270**.

(104) Additional underfill layers **730** may be disposed between the plurality of upper semiconductor chips **700**, and between the lower semiconductor chip **600** and the lowermost one of the plurality of upper semiconductor chips **700**. The additional underfill layer **730** may fill a space between the upper bumps **720**. The additional underfill layer **730** may include an insulating polymer material such as an epoxy resin.

(105) An additional mold layer **750** may be disposed on the first surface **600a** of the lower semiconductor chip **600** and may cover sidewalls of the plurality of upper semiconductor chips

700. In some embodiments, the additional mold layer **750** may include the same material as the mold layer **240**. In certain embodiments, the additional mold layer **750** may include a different material from that of the mold layer **240**. The additional mold layer **750** may include an insulating material (e.g., an epoxy molding compound).

(106) According to the present embodiments, the plurality of upper semiconductor chips **700** may be mounted on the first surface **600a** of the lower semiconductor chip **600**, and the mold layer **240** may be disposed on the second surface **600b** of the lower semiconductor chip **600**. The mold layer **240** may include the insulating material of which the coefficient of thermal expansion is greater than that of the lower semiconductor chip **600**, thereby reducing warpage of the lower semiconductor chip **600** which may be caused by structural imbalance between the lower semiconductor chip **600** and the plurality of upper semiconductor chips **700**. The mold layer **240** may include the fillers (e.g., silica), and thus the coefficient of thermal expansion of the mold layer **240** may be easily adjusted. As a result, the warpage of the lower semiconductor chip **600** may be easily adjusted.

(107) FIG. **21** is a cross-sectional view illustrating a semiconductor package **1700** according to some embodiments of the inventive concepts. Hereinafter, differences between the semiconductor package **1700** and the semiconductor package **1600** described with reference to FIG. **20** will be mainly described for the purpose of ease and convenience in explanation.

(108) Referring to FIG. **21**, the lower semiconductor chip **600** may include a lower interconnection layer **620** adjacent to the second surface **600b**. The lower interconnection layer **620** may include lower metal interconnection lines **622** adjacent to the second surface **600b**, and a lower interconnection insulating layer **623** covering the lower metal interconnection lines **622**. The lower semiconductor chip **600** may include a circuit layer (not shown) adjacent to the first surface **600a**.

(109) A plurality of first through-electrodes **250a** may be disposed in the lower semiconductor chip **600**. Each of the plurality of first through-electrodes **250a** may penetrate the lower semiconductor chip **600** so as to be connected to a corresponding one of the lower metal interconnection lines **622**. A plurality of second through-electrodes **250b** may be disposed in the mold layer **240**. Each of the plurality of second through-electrodes **250b** may penetrate the mold layer **240** so as to be connected to a corresponding one of the lower metal interconnection lines **622**. The lower interconnection layer **620** may be disposed between the plurality of first through-electrodes **250a** and the plurality of second through-electrodes **250b**. The plurality of first through-electrodes **250a** may be connected to the plurality of second through-electrodes **250b** through the lower interconnection layer **620**. Each of the plurality of first through-electrodes **250a** may vertically overlap a corresponding one of the plurality of second through-electrodes **250b**. However, the present invention is not limited thereto. In an exemplary embodiment, at least one of the plurality of second through-electrodes **250b** does not vertically overlap its corresponding one of the plurality of first through-electrodes **250a**.

(110) The plurality of first through-electrodes **250a** may be connected to the upper bumps **720**. The lowermost upper semiconductor chip **700** may be electrically connected to the lower semiconductor chip **600** via those connections between the plurality of first through-electrodes **250a** and the upper bumps **720**. The plurality of upper semiconductor chips **700** may be electrically connected to the lower substrate **100** through the through-vias **710**, the upper bumps **720**, the plurality of first through-electrodes **250a**, the lower metal interconnection lines **622**, the plurality of second through-electrodes **250b**, and the plurality of bumps **270**.

(111) Except for the differences as described above, other components and features of the semiconductor package **1700** according to the present embodiments may be substantially the same as corresponding components and features of the semiconductor package **1600** described with reference to FIG. **20**.

(112) According to the embodiments of the inventive concepts, the plurality of semiconductor chips may be mounted on the first surface of the upper substrate, and the mold layer may be disposed on

the second surface of the upper substrate. The mold layer may include the insulating material of which the coefficient of thermal expansion is greater than that of the upper substrate, and thus warpage of the semiconductor package may be reduced. The mold layer may include the fillers (e.g., silica), and thus the coefficient of thermal expansion of the mold layer may be easily adjusted. Therefore, the warpage of the semiconductor package may be easily adjusted.

(113) Furthermore, the buffer layer may be disposed on the second surface of the upper substrate, and the plurality of bumps may be disposed on the buffer layer so as to be spaced apart from the mold layer. The buffer layer may reduce stress applied to the plurality of bumps.

(114) As a result, the warpage may be easily controlled, and the semiconductor package with excellent reliability may be provided.

(115) While the inventive concepts have been described with reference to example embodiments, it will be apparent to those skilled in the art that various changes and modifications may be made without departing from the spirits and scopes of the inventive concepts. Therefore, it should be understood that the above embodiments are not limiting, but illustrative. Thus, the scopes of the inventive concepts are to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing description.

Claims

1. A semiconductor package comprising: a lower semiconductor chip having a first surface and a second surface which are opposite to each other; an upper semiconductor chip on the first surface of the lower semiconductor chip; a lower substrate on the second surface of the lower semiconductor chip; a mold layer between the second surface of the lower semiconductor chip and the lower substrate; a first through-electrode penetrating the lower semiconductor chip; a second through-electrode penetrating the mold layer; and a bump between the mold layer and the lower substrate, wherein the first through-electrode is connected to the second through-electrode, the second through-electrode is connected to the bump, and the bump is connected to the lower substrate, wherein the mold layer includes an insulating material of which a coefficient of thermal expansion is greater than that of the lower semiconductor chip, and wherein a thickness in a first direction of the mold layer is less than a thickness in the first direction of the lower semiconductor chip, and the first direction is perpendicular to the first surface of the lower semiconductor chip.
2. The semiconductor package of claim 1, wherein the first through-electrode and the second through-electrode overlap each other in the first direction perpendicular to the first surface of the lower semiconductor chip.
3. The semiconductor package of claim 2, wherein a first width in a second direction of the first through-electrode is less than a second width in the second direction of the second through-electrode, and the second direction is parallel to the first surface of the lower semiconductor chip.
4. The semiconductor package of claim 1, wherein the mold layer includes silica.
5. The semiconductor package of claim 1, further comprising: an upper bump between the first surface of the lower semiconductor chip and the upper semiconductor chip, wherein the first through-electrode is connected to the upper bump.
6. The semiconductor package of claim 5, further comprising: a through-via penetrating the upper semiconductor chip, wherein the through-via is connected to the upper bump.
7. The semiconductor package of claim 6, wherein the upper semiconductor chip is electrically connected to the lower semiconductor chip through the through-via, the upper bump and the first through-electrode.
8. The semiconductor package of claim 6, further comprising: an underfill layer between the first surface of the lower semiconductor chip and the upper semiconductor chip, wherein the underfill layer covers side surfaces of the upper bump.
9. The semiconductor package of claim 1, wherein the lower semiconductor chip is electrically

connected to the lower substrate through the first through-electrode, the second through-electrode and the bump.

10. The semiconductor package of claim 1, further comprising: a lower underfill layer between the mold layer and the lower substrate, wherein the lower underfill layer covers side surfaces of the bump.

11. A semiconductor package comprising: a lower semiconductor chip having a first surface of and a second surface which are opposite to each other, the lower semiconductor chip including a lower interconnection layer adjacent to the second surface, wherein the lower interconnection layer includes a plurality of lower interconnection lines; an upper semiconductor chip on the first surface of the lower semiconductor chip; a lower substrate on the second surface of the lower semiconductor chip; a mold layer between the second surface of the lower semiconductor chip and the lower substrate; a first through-electrode penetrating the lower semiconductor chip and connected to a corresponding one of the plurality of lower interconnection lines of the lower interconnection layer; a second through-electrode penetrating the mold layer and connected to a corresponding one of the plurality of lower interconnection lines of the lower interconnection layer; and a bump between the mold layer and the lower substrate, wherein the second through-electrode is connected to the bump, and the bump is connected to the lower substrate, wherein the first through-electrode is connected to the second through-electrode through the lower interconnection layer, and wherein the mold layer includes an insulating material of which a coefficient of thermal expansion is greater than that of the lower semiconductor chip.

12. The semiconductor package of claim 11, wherein a thickness in a first direction of the mold layer is less than a thickness in the first direction of the lower semiconductor chip, and the first direction is perpendicular to the first surface of the lower semiconductor chip.

13. The semiconductor package of claim 12, wherein a first width in a second direction of the first through-electrode is less than a second width in the second direction of the second through-electrode, and the second direction is parallel to the first surface of the lower semiconductor chip.

14. The semiconductor package of claim 11, wherein the lower interconnection layer further includes a lower interconnection insulating layer covering the plurality of lower interconnection lines.

15. The semiconductor package of claim 11, wherein the mold layer includes silica.

16. The semiconductor package of claim 11, further comprising: an upper bump between the first surface of the lower semiconductor chip and the upper semiconductor chip, wherein the first through-electrode is connected to the upper bump.

17. The semiconductor package of claim 16, further comprising: a through-via penetrating the upper semiconductor chip, wherein the through-via is connected to the upper bump.

18. The semiconductor package of claim 17, further comprising: an underfill layer between the first surface of the lower semiconductor chip and the upper semiconductor chip, wherein the underfill layer covers side surfaces of the upper bump.
